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Tugas Hand-Calculation Backpropagation

Rules

$$\frac{d}{du} e^u = e^u$$

$$\frac{d}{du}[f(u)g(u)] = f(u)g'(u) + f'(u)g(u) \Rightarrow \text{Chain Rule}$$

Logistic Function Derivative

$$\boxed{f(u) = \frac{1}{1+e^{-u}}} \Rightarrow \text{Logistic function}$$

$$= \frac{1}{1+\frac{1}{e^u}}$$

$$= \frac{1}{\frac{e^u+1}{e^u}}$$

$$\boxed{f(u) = \frac{e^u}{1+e^u}} \Rightarrow \text{Logistic function}$$

$$1-f(u) = \frac{(1+\cancel{e^u}) - \cancel{e^u}}{(1+e^u)}$$

$$\boxed{1-f(u) = \frac{1}{1+e^u}}$$

$$\frac{d}{du} f(u) = \frac{d}{du} e^u (1+e^u)^{-1}$$

$$= e^u(-1)(1+e^u)^{-2}(e^u) + e^u(1+e^u)^{-1}$$

$$= e^u(1+e^u)^{-1} - e^u \cdot e^u \cdot (1+e^u)^{-2}$$

$$= \frac{e^u}{(1+e^u)} - \frac{e^u \cdot e^u}{(1+e^u)^2}$$

$$= \frac{e^u(1+e^u) - e^u \cdot e^u}{(1+e^u)^2}$$

$$= \frac{e^u + \cancel{(e^u)^2} - \cancel{(e^u)^2}}{(1+e^u)^2}$$

$$= \frac{e^u}{(1+e^u)^2}$$

$$= \frac{e^u}{(1+e^u)(1+e^u)}$$

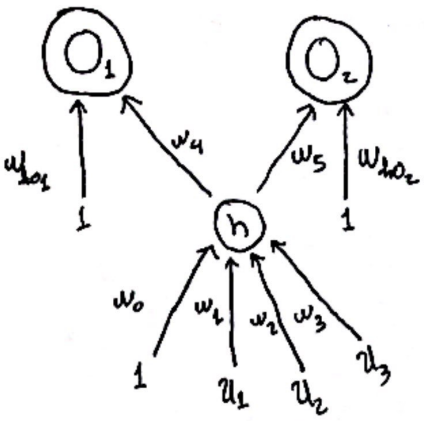
$$= \frac{f(u)}{(1+e^u)}$$

$$= f(u) \cdot \frac{1}{(1+e^u)}$$

$$\boxed{\frac{d}{du} f(u) = f(u)(1-f(u))} \Rightarrow \text{The derivative of Logistic Function}$$

Network

$$\boxed{E = \sum \frac{(t-o_2)^2}{2}} \Rightarrow \text{Error function}$$



Pencarian w_5

$$\frac{\partial}{\partial w_5} E_{total} = \frac{\partial E_{total}}{\partial O_2} \times \frac{\partial O_2}{\partial net_{o_2}} \times \frac{\partial net_{o_2}}{\partial w_5}$$

$$\Rightarrow \frac{\partial E_{total}}{\partial O_2} = \frac{\partial}{\partial O_2}\left[\frac{(t_2-O_2)^2}{2} + \frac{(t_1-O_1)^2}{2}\right]$$

$$= \left[\frac{\partial}{\partial O_2}\frac{(t_2-O_2)^2}{2}\right] + \left[\frac{\partial}{\partial O_2}\frac{(t_1-O_1)^2}{2}\right]$$

$$= \left[\cancel{2}\cdot\frac{1}{\cancel{2}}\cdot(t_2-O_2)\cdot(-1)\right] + 0$$

$$= (-1)(t_2-O_2)$$

$$\Rightarrow \frac{\partial O_2}{\partial net_{o_2}} = \frac{\partial}{\partial net_{o_2}}\frac{1}{1+e^{-net_{o_2}}}$$

$$= O_2(1-O_2)$$

$$\Rightarrow \frac{\partial net_{o_2}}{\partial w_5} = \frac{\partial}{dw_5}\left[h\cdot w_5 + 1\cdot w_{bo_2}\right]$$

$$= \left[\frac{\partial}{\partial w_5} h\cdot w_5\right] + \left[\frac{\partial}{\partial w_5} 1\cdot w_{bo}\right]$$

$$= \left[h\cdot 1\right] + 0 = h$$

$\Rightarrow$ sehingga

$$\boxed{\frac{\partial}{\partial w_5} E_{total} = (-1)(t_2-O_2) \times O_2(1-O_2) \times h}$$

Penurunan w_{h0_2}

$$\frac{\partial}{\partial w_{h0_2}} E_{total} = \frac{\partial E_{total}}{\partial O_2} \times \frac{\partial O_2}{\partial net_{O_2}} \times \frac{\partial net_{O_2}}{\partial w_{h0_2}}$$

$$\Rightarrow \frac{\partial net_{O_2}}{\partial w_{h0_2}} = \frac{\partial}{\partial w_{h0_2}} \left[h w_5 + 1 w_{h0_2} \right]$$

$$= \left[\frac{\partial}{\partial w_{h0_2}} h w_5 \right] + \left[\frac{\partial}{w_{h0_2}} 1 w_{h0_2} \right]$$

$$= 0 + [1]$$

$$= 1$$

$\Rightarrow$ sehingga

$$\boxed{\frac{\partial}{\partial w_{h0_2}} E_{total} = (-1)(t_2 - O_2) \times O_2 (1 - O_2)}$$

Penurunan w_4

$$\frac{\partial}{\partial w_4} E_{total} = \frac{\partial E_{total}}{\partial O_1} \times \frac{\partial O_1}{\partial net_{O_1}} \times \frac{\partial net_{O_1}}{\partial w_4}$$

$$\Rightarrow \frac{\partial}{\partial O_1} E_{total} = \frac{\partial}{\partial O_1} \left[\frac{(t_2 - O_2)^2}{2} + \frac{(t_1 - O_1)^2}{2} \right]$$

$$= \left[\frac{\partial}{\partial O_1} \frac{(t_2 - O_2)^2}{2} \right] + \left[\frac{\partial}{\partial O_1} \frac{(t_1 - O_1)^2}{2} \right]$$

$$= 0 + \left[\cancel{2} \cdot \frac{1}{\cancel{2}} (t_1 - O_1)(-1) \right]$$

$$= (-1)(t_1 - O_1)$$

$$\Rightarrow \frac{\partial}{\partial net_{O_1}} O_1 = \frac{\partial}{\partial net_{O_1}} \frac{1}{1 + e^{-net_{O_1}}}$$

$$= O_1 (1 - O_1)$$

$$\Rightarrow \frac{\partial}{\partial w_4} net_{O_1} = \frac{\partial}{\partial w_4} \left[h w_4 + 1 \cdot w_{h0_1} \right]$$

$$= h \cdot 1 + 0$$

$$= h$$

$\Rightarrow$ sehingga

$$\boxed{\frac{\partial E_{total}}{\partial w_4} = (-1)(t_1 - O_1) \times O_1 (1 - O_1) + h}$$

Pencarian w_{h0_1}

$$\frac{\partial E_{total}}{\partial w_{h0_1}} = \frac{\partial E_{total}}{\partial O_1} \times \frac{\partial O_1}{\partial net_{O_1}} \times \frac{\partial net_{O_1}}{\partial w_{h0_1}}$$

$$\Rightarrow \frac{\partial net_{O_1}}{\partial w_{h0_1}} = \frac{\partial}{\partial w_{h0_1}}\left[h w_4 + 1 w_{h0_1}\right]$$

$$= \left[\frac{\partial}{\partial w_{h0_1}} h w_4\right] + \left[\frac{\partial}{\partial w_{h0_1}} 1 w_{h0_1}\right]$$

$$= 0 + h.1$$

$$= h.$$

$\Rightarrow$ Sehingga

$$\boxed{\begin{aligned}\frac{\partial E_{total}}{\partial w_{h0_1}} &= \frac{\partial E_{total}}{\partial O_1} \times \frac{\partial O_1}{\partial net_{O_1}} \times \frac{\partial net_{O_1}}{\partial w_{h0_1}} \\ &= (-1)(t_1 - O_1) \times O_1(1-O_1) \times h\end{aligned}}$$

Pencarian w_3

$$\frac{\partial E_{total}}{\partial h} = \frac{\partial E_1}{\partial h} + \frac{\partial E_2}{\partial h}$$

$$\frac{\partial E_1}{\partial h} = \frac{\partial E_{total}}{\partial O_1} \times \frac{\partial O_1}{\partial net_{O_1}} \times \frac{\partial net_{O_1}}{\partial h}$$

$$\Rightarrow \frac{\partial net_{O_1}}{\partial h} = \frac{\partial}{\partial h}\left[h w_4 + 1 . w_{h0_1}\right]$$

$$= \left[\frac{\partial}{\partial h} h w_4\right] + \left[\frac{\partial}{\partial h} 1 . w_{h0_1}\right]$$

$$= (1 . w_4) + 0$$

$$= w_4.$$

$$\boxed{\frac{\partial E_1}{\partial h} = (-1)(t_1 - O_1) + O_1(1-O_1) + w_4} \quad \ldots\ldots (1)$$

$$\frac{\partial E_2}{\partial h} = \frac{\partial E_{total}}{\partial O_2} \times \frac{\partial O_2}{\partial net_{o_2}} \times \frac{\partial net_{o_2}}{\partial h}$$

$$\Rightarrow \frac{\partial net_{o_2}}{\partial h} = \frac{\partial}{\partial h}\left[h w_5 + 1 . w_{h o_2}\right]$$

$$= \left[\frac{\partial}{\partial h} h w_5\right] + \left[\frac{\partial}{\partial h} 1 . w_{h o_2}\right]$$

$$= (1 . w_5) + 0$$

$$= w_5$$

$$\boxed{\frac{\partial E_2}{\partial h} = (-1)(t_2 - O_2) + O_2(1 - O_2) + w_5} \quad \ldots\ldots\ldots (2)$$

$$\boxed{\frac{\partial E_{total}}{\partial h} = \left[(-1)(t_1 - O_1) + O_1(1 - O_1) + w_4\right] + \left[(-1)(t_2 - O_2) + O_2(1 - O_2) + w_5\right]} \quad \ldots (3)$$

$$\frac{\partial h}{\partial net_h} = \frac{\partial}{\partial net_h} \frac{1}{1 + e^{-net_h}}$$

$$= h(1 - h)$$

$$\frac{\partial net_h}{\partial w_3} = \frac{\partial}{\partial w_3}\left[1 . w_0 + u_1 w_1 + u_2 w_2 + u_3 w_3\right]$$

$$= \left[\frac{\partial}{\partial w_3} 1 . w_0\right] + \left[\frac{\partial}{\partial w_3} u_1 w_1\right] + \left[\frac{\partial}{\partial w_3} u_2 w_2\right] + \left[\frac{\partial}{\partial w_3} u_3 w_3\right]$$

$$= 0 + 0 + 0 + u_3$$

$$= u_3$$

$\Rightarrow$ Sehingga,

$$\frac{\partial E_{total}}{\partial w_3} = \frac{\partial E_{total}}{\partial h} \times \frac{\partial h}{\partial net_h} \times \frac{\partial net_h}{\partial w_3}$$

$$\boxed{\frac{\partial E_{total}}{\partial w_3} = \left(\left[(-1)(t_1 - O_1) \times O_1(1 - O_1) \times w_4\right] + \left[(-1)(t_2 - O_2) \times O_2(1 - O_2) \times w_5\right]\right) \times h(1 - h) \times u_3}$$

Pencarian w_2

$$\frac{\partial E_{total}}{\partial w_2} = \frac{\partial E_{total}}{\partial h} + \frac{\partial h}{\partial net_{h_1}} + \frac{\partial net_{h_1}}{\partial w_2}$$

$$\Rightarrow \frac{\partial net_{h_1}}{\partial w_2} = \frac{\partial}{\partial w_2}\left[1.w_0 + u_1 w_1 + u_2 w_2 + u_3 w_3\right]$$

$$= \left[\frac{\partial}{\partial w_2} 1.w_0\right] + \left[\frac{\partial}{\partial w_2} u_1 w_1\right] + \left[\frac{\partial}{\partial w_2} u_2 w_2\right] + \left[\frac{\partial}{\partial w_2} u_3 w_3\right]$$

$$= 0 + 0 + u_2 . 1 + 0$$

$$= u_2$$

$$\boxed{\frac{\partial E_{total}}{\partial w_2} = \left(\left[(-1)(t_1 - o_1) \times o_1(1-o_1) \times w_4\right] + \left[(-1)(t_2 - o_2) \times o_2(1-o_2) \times w_5\right]\right) \times h(1-h) \times u_2}$$

Pencarian w_1

$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial h} \times \frac{\partial h}{\partial net_{h_1}} \times \frac{\partial net_{h_1}}{\partial w_1}$$

$$\Rightarrow \frac{\partial net_{h_1}}{\partial w_1} = \frac{\partial}{\partial w_1}\left[1.w_0 + u_1 w_1 + u_2 w_2 + u_3 w_3\right]$$

$$= \left[\frac{\partial}{\partial w_1} 1.w_0\right] + \left[\frac{\partial}{\partial w_1} u_1 w_1\right] + \left[\frac{\partial}{\partial w_1} u_2 w_2\right] + \left[\frac{\partial}{\partial w_1} u_3 w_3\right]$$

$$= 0 + u_1 . 1 + 0 + 0$$

$$= u_1$$

$$\boxed{\frac{\partial E_{total}}{\partial w_1} = \left(\left[(-1)(t_1 - o_1) \times o_1(1-o_1) \times w_4\right] + \left[(-1)(t_2 - o_2) \times o_2(1-o_2) \times w_5\right]\right) \times h(1-h) \times u_1}$$

Pencarian w_0

$$\frac{\partial E_{total}}{\partial w_0} = \frac{\partial E_{total}}{\partial h} \times \frac{\partial h}{\partial net_h} \times \frac{\partial net_h}{\partial w_0}$$

$$\Rightarrow \frac{\partial net_h}{\partial w_0} = \frac{\partial}{\partial w_0}\left[1.w_0 + u_1 w_1 + u_2 w_2 + u_3 w_3\right]$$

$$= \left[\frac{\partial}{\partial w_0} 1.w_0\right] + \left[\frac{\partial}{\partial w_0} u_1 w_1\right] + \left[\frac{\partial}{\partial w_0} u_2 . w_2\right] + \left[\frac{\partial}{\partial w_0} u_3 w_3\right]$$

$$= 1.1 + 0 + 0 + 0$$

$$= 1$$

$$\boxed{\frac{\partial E_{total}}{\partial w_0} = \left(\left[(-1)(t_1 - o_1) \times o_1(1-o_1) \times w_4\right] + \left[(-1)(t_2 - o_2) \times o_2(1-o_2) \times w_5\right]\right) \times h(1-h) + 1}$$